

Applied Class 8 – Solutions

Part A

Early developmental experiences, such as incubation conditions, can have important consequences for post-hatching fitness in birds. A paper reported on a study where wood duck eggs were collected from nest boxes and experimentally incubated at various fixed temperatures, each falling within the range of temperatures of naturally incubated wood duck nests. The response variables recorded included egg mass (g) at the time of hatching. A one-way analysis of variance of egg mass by temperature in R gave the following results:

	Df	Sum Sq	Mean Sq	F value
Temperature	3	117.1	39.03	2.873
Residual	23	312.5	13.59	

- a) How many fixed temperatures were used in this design?

Since $DF_{\text{Temperature}} = 3 = k - 1$, there must have been 4 fixed temperatures

- b) How many eggs were included in this analysis?

$DF_{\text{Total}} = 3 + 23 = 26 = n - 1$, so there were 27 eggs.

- c) The paper reported that the mean egg mass for the 6 eggs at 35 °C was “44.7 ± 1.88 g” where they indicated that “1.88” was the standard error of the mean (SEM). Based on this statement, what was the sample standard deviation of the egg mass for those 6 eggs?

$$SEM = \frac{s}{\sqrt{n}}, \text{ so } 1.88 = \frac{s}{\sqrt{6}}, \text{ giving } s = \sqrt{6} \times 1.88 = 4.61 \text{ g}$$

- d) What is the R^2 value for this model?

$$R^2 = \frac{SS_{\text{Temp}}}{SS_{\text{Total}}} = \frac{117.1}{117.1 + 312.5} = 0.27$$

so incubation temperature explains 27% of the variability in egg mass.

- e) Complete the table to determine the F statistic used to test for the difference in mean egg mass between the different temperatures.

$$F = \frac{117.1/3}{312.5/23} = 2.873$$

- f) Use `pf()` in R to determine the p -value for this test. What do you conclude?

$P(F_{3,23} \geq 2.873) = 0.058$, using `1-pf(2.873, df1=3, df2=23)` in R. This gives only weak evidence of a difference in mean egg mass between the four incubation temperatures.

Part B

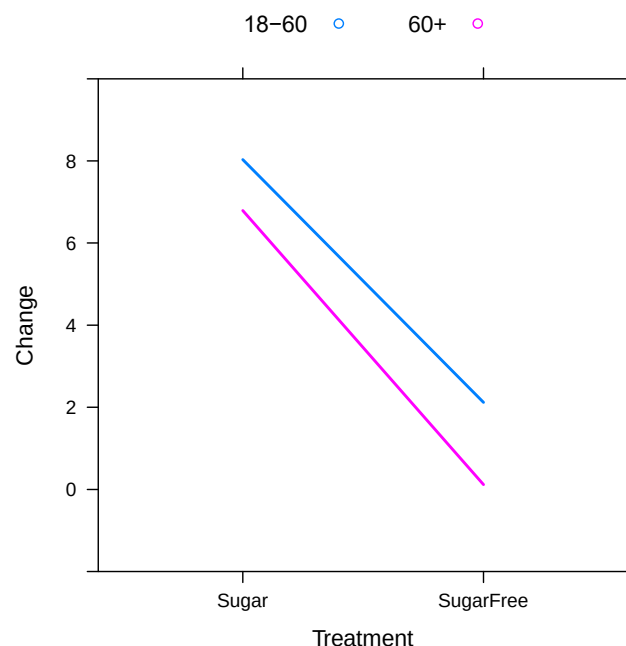
A project within the Islands investigated whether consuming sugar influences memory performance, and if any effects were the same for elderly subjects.

A sample of 40 subjects was randomly chosen from houses in Helluland, with 20 subjects aged 18-60 and 20 aged over 60. Each subject played a pairs memory game with 30 cards as quickly as they could, with the results recorded in seconds. Within each age group, half the subjects were randomly assigned to consume 50 g of lollies, while the other half consumed 50 g of sugar-free lollies. After 15 minutes, they repeated the memory game task.

- a) The table below shows the summary statistics for the decrease in memory game time between the first and second plays:

Age	Treatment	<i>n</i>	Mean (s)	SD (s)
18-60	Sugar	10	8.03	5.054
	Sugar-free	10	2.12	8.140
60+	Sugar	10	6.79	6.289
	Sugar-free	10	0.12	5.547

Draw an interaction effects plot for this data. Use the vertical axis for the mean change in game time and the horizontal axis for the treatment.



- b) Briefly discuss the potential interaction effect from your plot.

There seems to be a difference between the treatments and some difference between the age groups. However, since the lines are roughly parallel, there does not seem to be an interaction between the two factors in the effect on change.

- c) To compare the interaction effects of treatment and age group on the decrease in memory game time, a two-way analysis of variance is conducted, giving the following results in R:

```
summary(aov(Change ~ Treatment*Age, data=sugar))
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Treatment	1	395.6	395.6	9.761	0.00351 **
Age	1	26.2	26.2	0.647	0.42630
Treatment:Age	1	1.4	1.4	0.036	0.85135
Residuals	36	1459.2	40.5		

Fill in the table with the degrees of freedom for each of the variance components.

With four factor combinations, the Residuals degrees of freedom is $40 - 4 = 36$.

With two levels each, Treatment and Age will each have 1 degree of freedom, with the interaction then 1 as well.

Adding these together gives a total of 39 degrees of freedom, which matches $n - 1$.

- d) Calculate and interpret the R^2 value of this two-way ANOVA model.

$$R^2 = \frac{SS \text{ Model}}{SS \text{ Total}} = \frac{395.6 + 26.2 + 1.4}{395.6 + 26.2 + 1.4 + 1459.2} = 0.2248$$

so the combination of treatment and age group explains around 22% of the variability in the decrease in memory game time.

- e) Interpret the three p -values from the ANOVA table. What do you conclude about the effect of consuming sugar on memory performance, and how it relates to age?

There is strong evidence of a difference in mean change in memory game time between the sugar and sugar-free treatments ($F_{1,36} = 9.761, p = 0.00351$).

There is no evidence of a difference in mean change in memory game time between the age groups ($F_{1,36} = 0.647, p = 0.426$), and no evidence of an interaction between sugar treatment and age group on the mean change in memory game time ($F_{1,36} = 0.036, p = 0.851$).

Given the lack of evidence of an interaction, it may be useful to try an additive model, without the interaction term.

We encourage you to also use the `pf()` function in R to confirm the p -values directly.